AI-BASED CUSTOMER SUPPORT ASSISTANT USING NATURAL LANGUAGE PROCESSING AND REINFORCEMENT LEARNING

Datu, Marielle L.
Holy Angel University
D.Consunji St.
San Roque, Samal Bataan
+63 9991777947
marielledatu0611@gmail.com

Magtanong, Jasmine F.
Holy Angel University
Fatima Subdivision, Sapa Samal, Bataan
+63 9083964837
jasminemagtanong04@gmail.com

Martin, Masato
Holy Angel University
Villa Esperanza Lalawigan, Salama, Bataan
+63 9177564883
Masatomartin9@gmail.com

Salamat, Ivan Louis D.
Holy Angel University
Flores St. Mabatang, Bataan
+63 9672592061
Ivanlouis3@gmail.com

1. INTRODUCTION

Customer support is a critical component of business operations, directly influencing customer satisfaction, retention, and brand reputation [1]. With the proliferation of digital platforms, organizations now receive large volumes of customer inquiries through social media, messaging applications, and web portals. These inquiries often include repetitive questions about orders, account issues, product information, or service problems [2].

Manual handling of such requests requires significant human resources, which can lead to slow responses and inconsistent service quality. To address these challenges, artificial intelligence (AI) and natural language processing (NLP) technologies have been increasingly adopted in customer support systems [3]. NLP enables systems to interpret human language, classify user intent, and generate appropriate responses [4].

Deep learning models, particularly Convolutional Neural Networks (CNNs), have been applied to text classification tasks, providing robust performance in identifying semantic patterns within customer messages [5]. Additionally, Reinforcement Learning (RL) allows conversational systems to optimize response selection policies through iterative feedback and reward-based learning [6]. Combining CNN-based intent classification with RL-based response optimization enables automated assistants to adapt to customer interactions and improve over time [7].

This project proposes an AI-based Customer Support Assistant that leverages publicly available datasets of real customer-business interactions. The system will demonstrate practical applicability for business customer support, with a clear pathway for implementation within the project timeline.

2. PROBLEM STATEMENT AND SCOPE

Problem Statement

Businesses operating on digital platforms receive high volumes of customer inquiries, often involving repetitive and time-sensitive questions. Manual response handling can lead to delays, inconsistent service quality, and increased operational costs [1]. Traditional rule-based chatbots are limited by predefined rules and lack adaptability to diverse customer queries. Therefore, there is a need for a system that can automatically classify customer intents and select contextually appropriate responses while reducing the workload of human agents.

Scope

The proposed system will focus on:

- NLP-based preprocessing of customer messages (tokenization, embedding, normalization)
- CNN-based intent classification for user queries
- RL-based response selection to optimize interaction outcomes

- Evaluation of system performance on publicly available customer support datasets

Success Metrics

- **Accuracy and Macro-F1 Score** for intent classification
- **Task Success Rate** for automated responses
- **Cumulative Reward** for reinforcement learning policies
- **Response Latency** for end-to-end message handling

Constraints

- Models will be trained and evaluated on **publicly available datasets** [3]
- Implementation will focus on **single-turn customer queries**
- System design should support **efficient training and inference**, suitable for rapid prototyping

3. RELATED WORK

Recent studies have demonstrated the effectiveness of AI-powered customer support systems. Shum et al. [1] highlighted the evolution of conversational AI, emphasizing its role in automating routine customer interactions. Devlin et al. [2] demonstrated the use of transformer-based NLP models (BERT) for high-performance language understanding, enabling accurate intent classification in business contexts.

Zhang et al. [3] explored CNN architectures for sentence classification, showing superior performance over traditional machine learning methods in short-text classification tasks, such as customer support messages. Gao et al. [4] combined supervised learning with RL to optimize response strategies in dialogue systems, demonstrating improvements in task completion and user satisfaction.

Recent applications of RL in dialogue systems include contextual bandit models for policy optimization [5]. Sutton and Barto [6] provide the theoretical foundation for RL-based decision-making, enabling agents to learn optimal responses through interaction. Finally, real-world studies [7] have shown that leveraging large-scale social media datasets for training leads to more robust and context-aware customer support models.

4. DATASET PLAN

Primary Dataset

- **Customer Support on Twitter** [3]
- Contains >3 million public tweets with customer inquiries and business responses (Apple, Amazon, Uber, Spotify)
- Source: https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter

Governance

- License: Public research/educational use (Kaggle)
- Consent & Privacy: Pseudonymization of user information; public tweets only
- Representativeness: Diverse topics including orders, refunds, account issues, product inquiries

Preparation

- Remove usernames, URLs, and irrelevant text
- Tokenization, normalization, embedding conversion
- Split: 70% training, 15% validation, 15% testing

Additional Datasets

- **Customer Support Dataset (Multilingual)** – Kaggle https://www.kaggle.com/datasets/custsupport/multilingual
- **Dialog bAbI Tasks** (simplified task-oriented dialogues) https://research.fb.com/downloads/babi/

5. PLANNED CNN, NLP, AND RL COMPONENTS

NLP Component

- Preprocessing: tokenization, stopword removal, embedding with Word2Vec/GloVe
- Converts customer queries into numerical vectors for model input

CNN Component

- Embedding layer → convolutional layers (multiple filter sizes) → max pooling → dense layer → softmax output
- Classifies query intents into predefined categories

RL Component

- Contextual bandit agent maps predicted intents to candidate responses
- Reward function:
  - +1 for successful responses
  - 0 for neutral responses
  - −1 for incorrect responses
- Policy updated iteratively to maximize cumulative reward

Minimum Viable Product (MVP)

- Trained CNN intent classification model
- Response template database
- RL-based response selection policy
- Web or CLI interface for testing

6. ETHICS RISK REGISTER

Risk	Description	Mitigation
Incorrect automated responses	AI may provide misleading or inappropriate responses	Implement confidence thresholds and escalate uncertain cases to human agents
Privacy violations	Customer messages may contain sensitive info	Anonymize user data; only use public tweets; no PII stored
Data bias	Training data may overrepresent certain topics	Evaluate model across query types; consider oversampling underrepresented classes

7. TEAM ROLES AND MILESTONES

Team Roles

Member	Role
Marielle Datu	NLP preprocessing & documentation
Jasmine Magtanong	Data preprocessing & exploratory analysis
Masato Martin	CNN model design & training
Ivan Salamt	RL agent implementation & integration

Milestones

Week	Tasks
1	Proposal submission, GitHub repo setup, dataset acquisition
2	Data preprocessing, baseline model training
3	CNN + RL integration, model evaluation,

8. REFERENCES

[1] Shum, H., He, X., and Li, D. 2021. From Eliza to XiaoIce: Challenges and Opportunities with Social Chatbots. *arXiv preprint arXiv:2101.01360*. https://arxiv.org/abs/2101.01360

[2] Devlin, J., Chang, M., Lee, K., and Toutanova, K. 2021. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *arXiv preprint arXiv:1810.04805*. https://arxiv.org/abs/1810.04805

[3] Zhang, Y., Wallace, B., and Liu, S. 2022. A Sensitivity Analysis of Convolutional Neural Networks for Sentence Classification. *arXiv preprint arXiv:1510.03820*. https://arxiv.org/abs/1510.03820

[4] Gao, J., Galley, M., and Li, L. 2023. Neural Approaches to Conversational AI. *arXiv preprint arXiv:2106.04152*. https://arxiv.org/abs/2106.04152

[5] Kveton, B., Szepesvári, C., and Wen, Z. 2021. Contextual Bandits and Reinforcement Learning in Dialogue Systems. *Journal of Machine Learning Research*, 22(203), 1–32. http://jmlr.org/papers/v22/21-203.html

[6] Sutton, R. S., and Barto, A. G. 2021. *Reinforcement Learning: An Introduction* (2nd Edition). MIT Press. http://incompleteideas.net/book/the-book-2nd.html

[7] ThoughtVector. 2023. Customer Support on Twitter Dataset. *Kaggle Dataset*. https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter